

This is a comprehensive user manual for the Pro-Edge automatic edging and orgasm control device. Information and hardware/software downloads can be found in the devices GitHub Repository under:

github.com/jwgscott/pro-edge

Please report any bugs or errors you encounter via the GitHub “Issues” feature. This device is feature-complete for now – if you happen to come up with ideas for possible future changes or features, please do not hesitate to also open an “Issue” on GitHub.

1. Hardware & Interface Overview

The device is controlled via an interface consisting of two primary input/output components:

- **Rotary Encoder Knob with Push-button:** Used for navigating menus, selecting modes, and fine-tuning parameters (such as speed, sensitivity, and durations).
- **24-LED RGB Ring:** Provides real-time visual feedback regarding the current operating mode, motor speed, and pressure sensor telemetry at a 60Hz update rate.

Audio Feedback: The vibrator motor is utilized as a rudimentary speaker. It will emit distinct tones to signal power-on, power-off, and critical hardware warnings (such as the pressure sensor exceeding readable limits).

2. Power and Navigation Control

The push-button integrated into the encoder knob dictates the state machine of the device. The system distinguishes between three types of button presses based on the duration the button is held down.

Press Type	Duration	Primary Function
Short Press	< 600 ms	Cycles forward through the main operating modes or configuration menus.
Long Press	600 ms - 2500 ms	Jumps from a main operating mode into the associated Configuration Menu, or exits a Configuration Menu back to Manual mode.

Press Type	Duration	Primary Function
Very Long Press	> 2500 ms	Suspends device operation (Standby/Off). The LEDs and motor turn off, and a descending audio tone is played. The device remains off until the button is pressed again.

Note on Data Retention: User preferences, including maximum speed, sensitivity, target edges, and release duration etc. are persistently stored in the microcontroller's EEPROM. These settings are saved automatically when exiting the respective configuration menus and will survive a complete power cycle. These settings are lost on updating the device software.

NOTE: See “6. Initial Setup” for initial setup instructions.

3. Main Operating Modes

By utilizing a **Short Press**, the user cycles sequentially through the four primary operating modes. The current mode is easily identified by the color of the LED cursor on the ring.

3.1 Manual Control Mode (Red Cursor)

Indicator: Red LED cursor. Background LEDs display current pressure changes via a Green/Yellow/Red gradient.

Function: Direct, manual control over the vibrator's motor speed.

Use Case: Standard vibrator operation or baseline testing.

Knob Action: Turning the knob scales the motor PWM output linearly from 0 to the absolute maximum (255).

Possible settings configuration: None, fully manual mode.

3.2 Automatic Edging Mode (Pink Cursor/Blue Cursor)

Indicator: Pink or blue LED cursor in different hues – pink indicates release mode, blue indicates denial mode. Hues change over the adjustable edge detection sensitivity, keeping the general pink or blue theme. Background LEDs display current pressure changes via a Green/Yellow/Red gradient.

Release mode indicators:

- The **PROGRESS** is visualized by a white cursor moving forward towards an unmoving gold/yellow target cursor. Note: When variance is enabled, the progress bar still only informs about the set target without accounting for the variance, effectively lying to the user. If the variance is increasing the required number of edges for release, the white indicator will hit the gold target without granting release, keeping the user in *overtime*.
- An anti-cheat **PENALTY** (see below) is indicated by two red bars flashing quickly three times.
- **ACTIVE ORGASM** mode is indicated by the whole LED ring flashing brightly white.
- The **POST ORGASM** break is indicated by the whole LED ring slowly pulsing in a dark blue/purple color.

Function: The vibration motor ramps up to the configured maximum speed over the configured ramp up time. In abrupt mode, it will keep its defined speed until the defined sensitivity limit is reached/crossed.

In smooth mode, a proportional edging algorithm is used. Instead of cutting power instantly, the controller continuously calculates how close the current pressure delta is to the user-defined limit. As the pressure approaches the limit, the motor speed is dynamically and proportionally throttled down. If pressure/contraction exceeds the defined upper limit (set by the sensitivity), the motor shuts off abruptly (like in abrupt mode). After the set cool-down period, the ramp-up restarts.

If **DENIAL MODE** is selected, the device will keep executing this cycle until stopped by the user.

If **RELEASE MODE** is selected, the device will continue the cycle until a set number of edges is reached. (Set by the user, optional additional random variance unknown to the user) When the required number of edges is reached, the device will switch to the **ORGASM PHASE** upon the next pressure limit breach. Instead of switching off the vibrator, the vibration will continue for a set time before shutting off. This phase is indicated by the whole LED ring flashing brightly white. When the release phase concludes, a **POST ORGASM** break is started, indicated by the LED ring slowly pulsing in a dark blue/purple color. The length of this break can be defined by the user.

The post release break can be either set to none at all (continuing the edging cycle right after the orgasm phase, to a defined time or to end the automatic edging mode after reaching release, returning the device to manual mode right after.

When release mode is selected, the automatic edging mode utilizes an algorithmic **ANTI-CHEAT FUNCTION** to keep a user from raising the achieved edges counter by clenching rapidly or with a few second interval to deceive the device. This also keeps a single, strong edge with multiple clenches from being errone-

ously counted as multiple edges. For the next edge to be counted successfully (after reaching an initial edge), the user has to relax from clenching for an amount of time (a few seconds) – this resets the first lock. If the user keeps clenching continuously, the device will register a continuous edge, not resetting the detection and keeping the vibrator stopped.

After the user has successfully relaxed, the next edge has to be avoided for a random amount of time (shouldn't be too long to be incredibly difficult to avoid, especially in smooth mode) before edging again or a penalty is registered (indicated by two red bars flashing quickly three times). This means, that the edge (fake or real) isn't counted and the random timeout starts anew (with a new random duration). The device still registers the edge, switching of the vibrator and entering the forced relaxation mode to keep the penalty from triggering rapidly – only the edge isn't counted for progress to release.

Use Case: Automatic edging of the user based upon previously defined parameters and with an optional pre-defined release.

Knob Action: Adjusts the edge detection sensitivity. Turning the knob changes the acceptable pressure delta (limit) before a cutoff is triggered and how quickly the vibrator slows down in smooth mode. The sensitivity is indicated by the mode cursor, either in pink or blue hues. The full range of selectable sensitivity is indicated by the cursor doing three full circles around the LED ring – the higher the sensitivity, the lighter the mode color is indicated.

4. Configuration Menus

Configuration menus are accessed via a **Long Press** from the main operating modes.

4.1 Maximum Vibrator Speed (**Green Menu, static**)

Access: Long press from Manual or Automatic mode.

Indicator: Solid Green LEDs filled up to a white cursor indicating the maximum vibrator speed. Vibrator is active at the currently selected setting, giving direct feedback on changes.

Function: Sets the absolute maximum speed ceiling that the vibrator can reach in automatic edging mode. Is also used continuously during active orgasm mode. Should be set to the most pleasurable setting.

Knob Action: Adjusts the max speed ceiling.

Exit: A short press saves the newly selected maximum speed to the EEPROM memory and transitions the user into the ramp-up time (Green, 4.2) menu. A long press saves and exits the menu to the previously selected operating mode.

4.2 Ramp-Up Time Setting (Green Menu, ramping)

Access: Reached via a Short Press while currently inside the Maximum Motor Speed (Green, 4.1) menu.

Indicator: Green LED animation, ramping up to the white cursor, indicating the time the vibrator takes to ramp up to its full speed, both visually and physically by executing the selected ramp-up repeatedly.

Function: Defines the duration it takes for the vibrator motor to linearly accelerate from a stopped state up to the ceiling defined in the Maximum Motor Speed menu. This setting fundamentally dictates the pacing and urgency of the edging cycle in the automatic mode.

Knob Action: Adjusts the ramp-up duration. Turning the encoder maps the 24 LEDs to a time range spanning from 1 to 24 seconds, moving in 1-second increments.

Exit: A Short Press saves the newly selected post-edge break time to the EEPROM memory and transitions the user into the post-edge break (Dark Red, 4.3) menu. A long press saves and exits the menu to the previously selected operating mode.

4.3 Post-Edge break Setting (Dark Red Menu, pulsating)

Access: Reached via a Short Press while currently inside the ramp-up time (Green, 4.2) menu.

Indicator: Pulsating Dark Red LEDs filled up to a white cursor indicating the break time after an edge is registered.

Function: Sets the break time after an edge is registered until the vibrator starts up again and the next edging cycle starts.

Knob Action: Adjusts the post-edge break time. Turning the encoder maps the 24 LEDs to a time range spanning from 1 to 24 seconds, moving in 1-second increments.

Exit: A short press saves the newly selected post-edge break time to the EEPROM memory and transitions the user into the edging mode (abrupt/smooth) (Orange/Blue, binary) menu. A long press saves and exits the menu to the previously selected operating mode.

4.4 Abrupt/Smooth mode Setting (Orange/Blue binary)

Access: Reached via a Short Press while currently inside the post-edge break (Dark Red, 4.3) menu.

Indicator: Half of the LED ring is filled up either orange or blue while the other remains dark. The halves switch position and color depending on the setting.

Function: Sets the vibrator mode for the automatic edging mode. Abrupt mode switches the vibrator off abruptly after reaching the defined edge sensitivity limit. Smooth mode acts the same upon reaching the limit but also proportionally lowers the vibration intensity the closer the user gets to the set limit, increasing the intensity again dynamically after the pressure/arousal has dropped off.

Knob Action: Switches between abrupt and smooth mode. The encoder is set to be slightly sticky between settings, requiring three encoder clicks before switching between modes. Abrupt mode is indicated Orange, Smooth mode is indicated Blue.

Exit: A short press saves the newly selected edging mode (abrupt/smooth) to the EEPROM memory and transitions the user into the release/denial mode (Light Grey/Pink, binary) menu. A long press saves and exits the menu to the previously selected operating mode.

4.5 Denial/Release mode Setting (Light Grey/Pink binary)

Access: Reached via a Short Press while currently inside the edging mode (abrupt/smooth) (Orange/Blue, binary, 4.4) menu

Indicator: Half of the LED ring is filled up either light grey or pink while the other remains dark. The halves switch position and color depending on the setting.

Function: Sets the release mode for the automatic edging mode. Denial mode (light grey) continues the edging cycle indefinitely without granting release until switched off manually. Release mode (pink) grants a release after a user defined number of required edges. Enables additional release mode specific settings.

Knob Action: Switches between denial and release mode. The encoder is set to be slightly sticky between settings, requiring three encoder clicks before switching between modes. Denial mode is indicated Light Grey, Release mode is indicated Pink.

Exit: A short press saves the newly selected mode (denial/release) to the EEPROM memory and transitions the user into the next configuration menu depending on setting (denial/release) – see Note below. A long press saves and exits the menu to the previously selected operating mode.

NOTE: When Denial Mode is selected, the next menu after 4.5 (Denial/Release mode setting) is the Pressure Debug menu (4.10). The release mode specific settings are skipped. To exit the configuration menus and return to the main operating modes, do a long press.

When Release Mode is selected, menus specific to this mode become available. They are entered right after exiting (short press) the Denial/Release mode setting. Next setting is 4.6, Target Edges Setting. After the last Release mode specific menu, the user reaches the Pressure Debug menu like in Denial Mode. To exit the configuration menus and return to the main operating modes, do a long press.

4.6 Target Edges Setting (Magenta Menu, solid)

Access: Reached via a Short Press while currently inside the Denial/Release mode (Light Grey/Pink binary, 4.5) menu – only if Release (pink) mode is selected!

Indicator: Solid Magenta LEDs filled up to a white cursor.

Function: Defines how many edges the user must achieve in Auto-Release mode before the final release is triggered and the user reaches active orgasm mode on the next detected edge.

Knob Action: Selects the required target edge count in release mode, ranging from 2 to 48 edges in increments of 2 (every LED indicates two required edges).

Exit: A short press saves the newly selected amount of required edges to the EEPROM memory and transitions the user into the target edge variance (Cyan, 4.7) menu. A long press saves and exits the menu to the previously selected operating mode.

4.7 Target Edge Variance Setting (Cyan/Light Blue Menu, solid)

Access: Reached via a Short Press while currently inside the target edges (Magenta, 4.6) menu.

Indicator: Solid Cyan LEDs, filling blocks of 4, up to a white cursor.

Function: Introduces a dynamic, random element to the release mode to prevent predictability. This menu dictates a randomized +/- variance that is secretly applied to the baseline target edge count (set in 4.6) at the start of every release cycle. For example, if the baseline target is set to 12, and the variance is set to

2, the device will secretly require anywhere between 10 and 14 edges before triggering the release.

If activated, the white progress cursor in automatic edging mode intentionally lies to the user. It only displays progress relative to the baseline target. If the randomized variance selects a target higher than the baseline (set in 4.6), the white cursor will reach the gold target LED and stop. The user will be held in "overtime," forced to continue edging without visual feedback until the hidden target is finally met. If the randomized variance selects a target lower than the baseline, orgasm mode will be triggered early, before the cursor has reached the required edges. Variance can also randomly choose the user defined amount of required edges, not changing anything and triggering active orgasm mode right on target.

Note: The software prevents the randomized target from ever dropping below 1. A release phase cannot be triggered instantly upon starting a session. Please also take care to set a realistic variance depending on the set number of target edges for an enjoyable experience. (Probably: high variance only with a high number of required edges selected.)

Knob Action: Adjusts the variance range. Turning the encoder maps the random variance range from 0 to 6, visually indicated by the position of the White cursor over the Cyan LEDs. Setting this to 0 completely disables the random variance feature, ensuring the release triggers exactly on the configured number of target edges every time.

Exit: A short press saves the newly selected random variance range to the EEPROM memory and transitions the user into the release duration menu (Yellow, 4.8) menu. A long press saves and exits the menu to the previously selected operating mode.

4.8 Release Duration Setting (Gold/Yellow Menu, solid)

Access: Reached via a Short Press while currently inside the target edge variance (Cyan, 4.7) menu.

Indicator: Solid Yellow LEDs filled up to a white cursor with 1-3 LEDs.

Function: Defines the length of the Active Orgasm Phase in automatic edging mode with release enabled. In active orgasm phase, triggered directly upon registering an edge when the required number of edges is reached (including variance), the vibrator will be held on the maximum configured vibrator speed (see 4.1) for the duration set in this menu before switching off the vibrator completely and entering the post-orgasm break (see 4.9).

Changing this setting allows to drastically alter the release experience the user receives: Setting this duration intentionally short (0-6 seconds, depending on the user) almost inevitably results in a ruined orgasm while setting it to a normal

timeframe (about 8-24 seconds, depending on the user) grants a full, enjoyable orgasm. Extending the release duration beyond a certain threshold (depending on the user) results in post-orgasm torture, overstimulating the user.

Manipulating this setting is most effective when the user cannot interfere with the devices actions – using bondage for example.

Knob Action: Selects the duration of the vibration in active orgasm mode, ranging from 2 to 144 seconds in increments of 2. The setting allows the cursor to make three full revolutions around the LED ring, adding an LED to the cursor with each revolution: first revolution 2-48 seconds, cursor is one white LED. Second revolution: 50-96 seconds, cursor is two white LEDs. Third revolution: 98-144 seconds, cursor is three white LEDs.

Exit: A short press saves the newly selected release duration to the EEPROM memory and transitions the user into the post-orgasm break menu (Purple, 4.9) menu. A long press saves and exits the menu to the previously selected operating mode.

4.9 Post-Orgasm Break Setting (Purple Menu, pulsating)

Access: Reached via a Short Press while currently inside the release duration (Yellow, 4.8) menu.

Indicator: Pulsating Purple LEDs filled up to a white cursor.

Function: Defines the length of the post-orgasm break the device grants the user. Depending on setting, the break can either be non-existent, directly returning to the automatic edging cycle after the active orgasm phase has concluded (see 4.8), or it will pause the automatic edging cycle for a defined rest period before returning. This setting also allows the automatic edging cycle to end upon achieving release, ending the automatic edging mode and returning the user to the manual device control mode.

Knob Action: Selects the duration of the post-orgasm break in release mode, ranging from 0 to 57 seconds in increments of 3. Each step of the white cursor indicates an additional three seconds of break. The cursor at starting position without any purple LEDs visible disables the post-orgasm break completely, returning directly to the automatic edging cycle.

When filling up the LED Ring completely, four red LEDs at the end of the indicator will light up at the same time. This means the automatic edging cycle will end completely after completing the active orgasm phase, returning the user to manual device control mode.

Exit: A short press from this menu transitions into the pressure debug monitor (White, 4.10) menu, or a long press saves and exits the menu to the previously selected operating mode.

4.10 Pressure Debug Monitor (White Menu, solid)

Access: Short press from the either the automatic edging mode selector menu (4.5) when Denial mode is selected, or from the post-orgasm break menu (4.9) when Release mode is selected in 4.5.

Indicator: Solid (slightly dimmed) White LEDs filled up to a white cursor.

Function: A diagnostic tool that maps the raw analog voltage from the pressure sensor directly to the LED ring. The motor is forced off in this mode.

Use Case: Verifying that the inflatable plug is registering pressure correctly or troubleshooting hardware sensor sensitivity. Use this menu to adjust the hardware resistor for the pressure sensor. The indicated pressure should (usually) be roughly somewhere in the middle (or slightly lower than the middle of the LED Ring when the plug is inflated but in a resting state. If overvoltage/over-pressure is registered, the ring will be fully filled and the vibrator will produce a continuous beeping sound.

Exit: A short press transitions the user into the maximum vibrator speed menu (Green, 4.1) menu. A long press exits the menu to the previously selected operating mode.

5. Initial Setup

For initial setup, make sure the device is connected properly. The device needs a stable 12V power supply via the Arduinos input (not USB) for proper function of the vibrator. Connect the pressure hose with plug and inflator bulb.

After plugging in the power supply, the device should make an audible startup chime using the vibrator motor. If you cannot hear anything after 10-15 seconds, do a short press on the encoder knobs push button. The device should turn on in manual mode, indicating a red LED and (maybe) a green pressure indicator. If the device hasn't made any sound using the vibrators motor, the sound generation does not function properly. This should be noted as all other sound related functions will be unavailable.

When in manual mode, enter settings via a long press. The green vibrator maximum speed settings menu should appear (4.1) Do short presses until you reach the white pressure debug menu (4.10). Note the pressure indication. The indicated pressure should (usually) be roughly somewhere in the middle (or slightly lower than the middle of the LED Ring when the plug is inflated but in a resting state and near zero pressure indication if the plug is empty. If overvoltage/over-pressure is registered, the ring will be fully filled and the vibrator will produce a continuous beeping sound. If your device indicates an overpressure event, reduce the pressure sensors output voltage by adjusting the hardware potentiometric resistor until an acceptable input reading is achieved.

When the pressure is indicated at a maximum of one quarter of the pressure indicator, inflate the plug slightly (2-3 pumps). Now note the pressure readout again: it should read somewhere in the middle (or slightly lower than the middle of the LED Ring when the plug is slightly inflated. When squeezing the plug with your hand (slightly), a pressure increase of around 4-8 LEDs should be visible. If the pressure doesn't read at around half a ring of indication, adjust the potentiometer accordingly.

When adjusted properly, the device is ready to use. Adjust the settings according to your preferences using this guide and enjoy.

6. Technical Safety & Troubleshooting

- **Pressure Sensor Railing:** If the internal analog pressure reading exceeds a value of 4030, the system detects a sensor overload. The motor will emit three high-pitched warning beeps. If left unchecked, this condition leads to missed edge detections. To resolve this, adjust the hardware trimpot on the circuit board to lower the baseline analog gain.
- **Hardware Diagnostics:** The device streams real-time telemetry (Target Motor Speed, Raw Pressure, and 25-Second Running Average Pressure) over the USB Serial interface at a baud rate of 115200. This data can be monitored via the Arduino IDE Serial Monitor for advanced fine-tuning.
- **Thermal Management:** The software limits the overall LED brightness to a value of 50 to restrict current draw. Modifying the source code to display all LEDs at maximum white brightness (RGB 255,255,255) simultaneously is strictly prohibited, as it may overheat the voltage regulator and cause unexpected microcontroller resets.

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